

COLUMN | NIPUN MARYA

Future smartphones

Gene Roddenberry, who made Captain Kirk talk on the communicator in Star Trek, was not even a scientist, but he did inspire Martin Cooper to create the first mobile phone in 1973. Neither Roddenberry nor Cooper would have imagined where we would be close to 50 years hence. Looking 20 years into the future of smartphones, we have to look at a bit of science fiction and combine that with current advances in tech to know what smartphones of the future might offer.

Networks are constantly getting faster, more reliable, and ever pervasive. India had 3.6 million mobile phone internet users when 4G launched in 2012, which will increase to at least 500 million by 2023, post the probable launch of 5G in India. 4G has provided use cases and sparked businesses that could not be imagined with 3G, with content and video being examples. 5G promises more – with IoT and Augmented Reality being touted as some of the top applications that it would power. Over the next 20 years, we can expect two more generations of networks, focused on the use cases that emerge after we implement 5G. For example, forget about downloading one movie in a few minutes, with 6G speeds, you could potentially download 140 hours of movies in a second!

Smartphones of the future may not be smartphones at all. They will take on a whole new form factor that will enable us to perform the same tasks as smartphones do today – and much more. We have already witnessed multiple demonstrations of foldable phones attracting eyeballs, and they're only going to get more popular as more devices are released and they become more affordable. The next technological breakthrough in this area that you can envision is stretchable phones. This would allow you to quickly pull from the corners to increase, say, a 6-inch screen to 9-inches when watching videos, and then make it smaller

again to fit in your pocket, sort of like a rubber band. This will add a whole new dimension to gaming and video content on the phone.

Next we will have Morphed Phones. With the advent of more malleable materials and smaller electronics, it will soon be possible to create handsets that can be morphed to help you with whatever you are doing. If you are watching a video, they will be normal rectangular phones. But you could also bend them to wrap them around your wrist while jogging, or wear them as glasses that offer augmented reality. Although, technology is not quite there yet, it could happen with just a few breakthroughs in research.

This is more like fiction, but theoretically, your phones could just vanish. With the improvement in voice assistants, wearable devices, and faster networks like 5G, you might not even need a device! With physical smartphones, you are essentially interacting with people, networks, and devices using its screen. With devices like Smart Glasses, TWS, and smart-watches, the actual need for a physical device could become redundant, at least for a few use cases.

One of the biggest differentiating features in smartphones nowadays is the front-facing camera – selfies are one of the main types of pictures you take on your phones. Instead of big notches, teardrop notches, punch holes and pop-up selfie cameras, in the very near future, the front camera will actually be under the display. While camera innovations have already brought smartphones on par with DSLRs, I wouldn't be surprised to see a mini-detachable DSLR in smartphones.

Recognized worldwide for its selfie-shooting smartphones, Vivo is now readjusting its focus with ZEISS to integrate professional photography that suits the lifestyle of tech-savvy users.

It is argued that we're not surrounded by technology, we are



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consumed by it. We at Vivo, amongst others, have been working hard at providing information and tools to help our users balance their time with phones. While the first few years of the content explosion were about providing choice to the user – the next few will be about managing this choice effectively and ensuring that there is a balance between life and technology.

Last, but not least, healthcare will probably see the biggest leap of all as the speed and depth of connectivity improves. Also, with more and more sensors being developed to be a part of wearables and smartphones, we will see this combination saving many lives. It is bound to improve early detection of disease, better emergency response times, and improve healthcare quality, even in remote locations. With the current pandemic, this is the one use case that we are most excited about. The world is bound to focus more on providing improved medical facilities, and a good part of that could reside in the device in your pocket or on your wrist.

We are at an interesting inflection point in our relationship with technology and smartphones. This is the period which will mark the beginning of machine-to-machine communication. It will be interesting to see how this relationship co-exists with the relationship that machines have with us. The smartphone, at least for a while, is going to be a bridge between these two worlds, connecting them with each other. **4**